Supplementary Material for

EIT to CT Cross-Modality Translation Using Diffusion Transformer

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VQ-VAE Diagram and Architecture Details

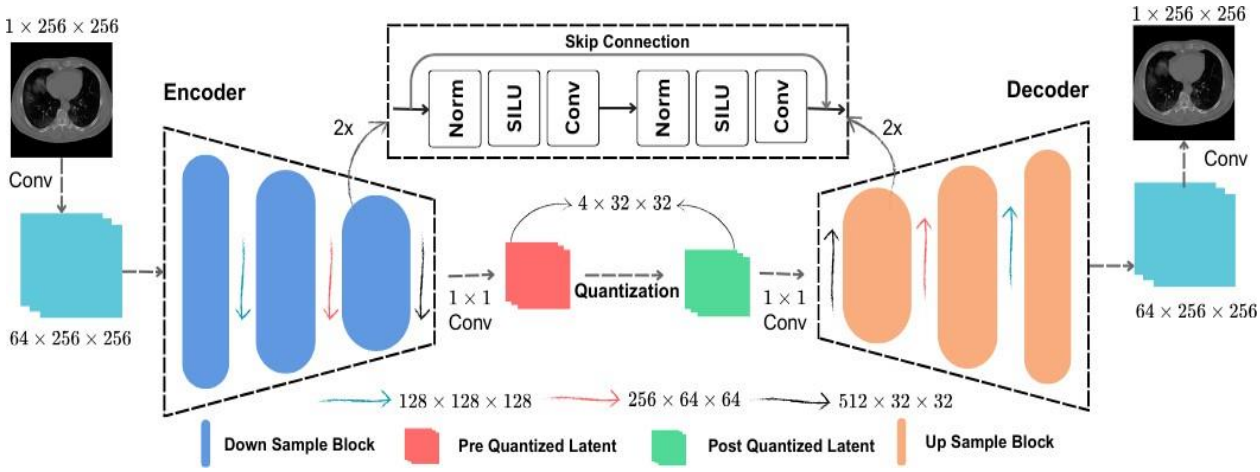

Fig: Architecture of the VQ-VAE model. Both the encoder and decoder consist of three blocks, with each encoder block followed by adownsampling layer and each decoder block by an upsampling layer. Each block includes two residual layers, each comprising two convolutional layers preceded by group normalization and SiLU activation

- A skip connection after the second convolution is added to form a residual path. At the end of each block, spatial downsampling is performed using a convolution layer with a stride of 2, resulting in a final encoder output of size 512 × 32 × 32.
- The decoder architecture mirrors the encoder in reverse, employing upsampling operations to progressively restore spatial resolution to 64 × 256 × 256, with a final convolutional layer producing the output image (1×256×256).
- The discriminator network processes input images through channels [64, 128, 256,1] using 4 × 4 kernels with stride 2, ultimately generating a patch-based authenticity map.

Training and Sampling Algorithms

The complete training procedure of the translation framework is outlined in Algorithm 1:

Algorithm 1 Training Approach for Conditional LDM in EIT-to-CT Translation

Input: Paired training data: (V, X) where X is a CT image and V the associated voltage measurements
Given: Pretrained VQ-VAE encoder E_{VQ}
Hyperparameters: Noise schedule $\{\beta_1, \beta_2, \ldots, \beta_T\}$; model parameters θ
Initialize Model parameters θ randomly
repeat
    Sample noise $\epsilon \sim \mathcal{N}(0, I)$
    Sample timestep $t \sim \mathcal{U}(1, T)$
    Encode CT: $z_0 = E_{\text{VQ}}(X)$
    Add noise: $z_t = \sqrt{\bar{\alpha}_t} \cdot z_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$
    Compute loss gradient: $\nabla_\theta \|\epsilon - D_\theta(z_t, V, t)\|^2$
    Update θ using Adam optimizer
until training convergence
Output: Trained diffusion model D_θ

While algorithm 2 represents the sampling process:

Algorithm 2 Sampling Method for Conditional LDM in EIT-to-CT Translation

Input: Voltage data V
Given: Trained diffusion model D_θ, VQ-VAE decoder D_{VQ}
Hyperparameters: DDIM schedule $\{\bar{\alpha}_1, \ldots, \bar{\alpha}_{50}\}$
Initialize latent: $z_{50} \sim \mathcal{N}(0, I)$
for $t = 50$ **to** 1 **do**
    Predict noise: $\epsilon_\theta = D_\theta(z_t, V, t)$
    Predict clean latent:

$$\hat{z}_0 = \frac{z_t - \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon_\theta}{\sqrt{\bar{\alpha}_t}}$$

    if $t > 1$ **then**
        Update latent:

$$z_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \cdot \hat{z}_0 + \sqrt{1 - \bar{\alpha}_{t-1}} \cdot \epsilon_\theta$$

    else
        Set $z_0 = \hat{z}_0$
    end if
end for
Decode CT image: $\hat{X} = D_{\text{VQ}}(z_0)$
Output: Generated sCT image $\hat{X}$

Ablation Studies on the Representation Learning Stage

Loss Functions

An ablation study was conducted on the representation learning stage to evaluate the contribution of individual components in the composite loss function. The loss function was systematically augmented, starting from MSE only, then successively incorporating adversarial and perceptual losses, and finally combining all three. The results demonstrated a clear, progressive improvement in reconstruction quality with the addition of complementary objectives (Table 1). The baseline model, trained with only the MSE loss, produced blurry reconstructions and achieved FID of 105.67 and KID of 0.1101, which were moderately improved by incorporating the adversarial loss (FID 91.85, KID 0.0871). However, the addition of perceptual loss to the MSE yielded dramatic gains, achieving FID of 24.23 and KID of 0.0073, underscoring its critical role in perceptual quality. The composite loss combining all three losses further refined reconstruction quality with FID of 21.16 and KID of 0.0033, thus validating the choice of a multi-objective function for learning perceptually realistic and structurally accurate image representations.

Latent Size Dimensionality

In order to determine the optimal dimensionality of the latent space, we varied the latent size in [16,32,64,128] and evaluated FID and KID scores. The results, demonstrated in Table 1, show that image quality improved monotonically but with diminishing returns. The transition from 16 to 32 provided the most substantial gain, accounting for approximately 74% of the total FID improvement and 55% of the total KID improvement achieved across the entire size range. Further increases to 64 and 128 yielded only marginal performance gains. Consequently, a latent size of 32 was selected for the proposed model, as it captures the majority of the representational capacity required for high-fidelity synthesis, while maintaining a more compact and computationally efficient latent space.

Ablation Study	Configuration	FID ↓	KID ↓
Loss Function	MSE Only	105.70	0.1101
	MSE + Adv.	91.81	0.0871
	MSE + LPIPS	24.22	0.0073
	MSE + LPIPS + Adv	20.50	0.0031
Latent Size	16	21.42	0.0042
	32	20.50	0.0031
	64	20.32	0.0027
	128	20.21	0.0022

Tab 1: Analysis of loss functions and latent size performed on the representation learning stage of the proposed method

Ablation Study on Sensitivity to Contact Impedance

We evaluated the model's robustness to uncertainties in contact impedance, by perturbing its value across a wide range (0.0025 to 0.1). Table 2 shows that the reconstruction quality remained largely stable, with quantitative metrics showing minimal fluctuations. At the largest deviation, PSNR varied by less than 0.1 dB, while FSIM and CC changed by less than 0.2% and 0.1%, respectively. Perceptual metrics showed only minimal degradation of approximately 0.6-0.7%. Distributional metrics were more sensitive, with FID increasing by 0.93 (2.8%) and KID by 0.0013 (11.82%) at the largest impedance deviation. This indicates that the proposed EIT-to-CT translation method remains robust to realistic variability in electrode–skin contact impedance, a common challenge in EIT measurements.

Impedance	CC ↑	PSNR ↑	FSIM ↑	LPIPS ↓	DISTS ↓	FID ↓	KID ↓
0.0025	0.9279	24.58	0.8305	0.1913	0.1288	33.47	0.0113
0.005	0.9282	24.60	0.8314	0.1914	0.1284	33.39	0.0112
Actual (0.01)	0.9282	24.64	0.8321	0.1916	0.1289	33.36	0.0110
0.02	0.9283	24.65	0.8324	0.1912	0.1286	33.86	0.0116
0.05	0.9278	24.66	0.8326	0.1922	0.1289	34.12	0.0119
0.08	0.9276	24.62	0.8322	0.1927	0.1298	34.26	0.0121
0.10	0.9275	24.62	0.8322	0.1927	0.1297	34.29	0.0123

Tab 2: Model robustness to contact impedance change errors in terms of image reconstruction quality.

Quantitative Metrics Detail

CC determines the strength and direction of the linear relationship between the pixels of two images. For an actual image *X* and a generated image *Y*, CC is given by::

$$CC = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

where σ_x, σ_Y, and σ_{xY} represent the standard deviations and covariance of the images.

PSNR estimates the pixel-wise fidelity between the actual and generated images by measuring the ratio between the maximum pixel value X_{max} and the MSE introduced during the generation process:

$$\text{PSNR} = 10 \cdot \log_{10}\left(\frac{X_{max}^2}{\text{MSE}}\right)$$

FSIM evaluates perceptual similarity by modeling features that align with visual perception, such as Phase Congruency (PC) and Gradient Magnitude (GM). It is defined as:

$$\text{FSIM} = \frac{\sum_i S_L(x) \cdot PC_m(x)}{\sum_i PC_m(x)}$$

where *i* is the pixel index, *S_L(x)* is the local similarity, and *PC_m(x)* is the maximum phase congruency.

LPIPS evaluates perceptual similarity between images by estimating the weighted sum of the difference between their normalized deep features from the layers of a pretrained network *f*. It is calculated as:

$$\text{LPIPS}(\mathbf{X}, \mathbf{Y}) = \sum_l w_l \cdot \|\hat{\mathbf{f}}_l(\mathbf{X}) - \hat{\mathbf{f}}_l(\mathbf{Y})\|_2^2$$

where $\hat{f}_l$ represents channel-wise normalized features from layer *l* and w_l are learned weights that reflect the perceptual importance of each channel.

DISTS extends this idea by modeling both structural and textural information from the deep features to better represent visual similarity. This is defined as:

$$\text{DISTS}(\mathbf{X}, \mathbf{Y}) = \sum_l \alpha_l \cdot S_l(\mathbf{X}, \mathbf{Y}) + \beta_l \cdot T_l(\mathbf{X}, \mathbf{Y})$$

where α_l and β_l are weights, and S_l and T_l represent structural and textural components, respectively.

FID measures the Fr'echet distance between the Gaussian distributions of the feature representations (typically using InceptionV3) for the generated and real images, comparing their means and covariances:

$$\text{FID} = \|\boldsymbol{\mu}_r - \boldsymbol{\mu}_g\|_2^2 + \text{Tr}\left(\boldsymbol{\Sigma}_r + \boldsymbol{\Sigma}_g - 2\left(\boldsymbol{\Sigma}_X \boldsymbol{\Sigma}_g\right)^{1/2}\right)$$

where Tr is the trace operator, μ_r and μg are means, and Σ_r and Σg are covariances..

KID is similar to FID but uses polynomial kernel comparisons, specifically, the squared Maximum Mean Discrepancy (MMD), between feature distributions F_r and F_g. It is defined as:

$$\text{KID} = \text{MMD}^2(\mathcal{F}_r, \mathcal{F}_g)$$